Cross-species Unified ncRNA Inference and Annotation

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Abstract

Reference-based annotation transfer via orthology has been highly successful for protein-coding genes but remains challenging for non-coding RNAs (ncRNAs) due to weak sequence conservation, ambiguous transcript boundaries, and incomplete annotations. Here I present CURIA, a pipeline for cross-species ncRNA annotation that combines genome alignment chains with embedding-based representations derived from RNA foundation models. CURIA formulates ncRNA orthology inference as synteny-constrained matching in embedding space. It first identifies candidate orthologous loci using a simplified TOGA-based projection, then detects localized structured subregions ("islands") within transcripts, and finally establishes cross-species correspondence using distribution-level similarity of embedding representations. Using a conservative annotation-based evaluation across eight mammalian genomes, CURIA recovered 53.1% of predicted short ncRNA loci with annotation support and identified ~700 candidate conserved intergenic RNA-like loci not present in current annotations. These results show that combining synteny with embedding-based similarity can support orthology-aware ncRNA annotation beyond sequence conservation, establishing a proof of concept for combining comparative genomics with RNA foundation-model embeddings for genome-scale ncRNA annotation.

1. Introduction

1.1 Protein-coding orthology as a tractable problem

Orthology inference for protein-coding genes is well established and supported by scalable annotation transfer pipelines[1–7]. Strong sequence conservation, together with constraints imposed by open reading frames, enables reliable cross-species mapping of gene loci[2]. Methods such as TOGA, which combine genome alignment chains with coding sequence structure, achieve high accuracy at genome scale. However, these approaches rely on properties specific to protein-coding genes and do not generalize to non-coding RNAs. In particular, they assume conserved coding structure and detectable sequence similarity, which are often absent outside protein-coding contexts. In this work, we do not attempt to infer evolutionary orthology in the strict sense. Instead, we focus on identifying candidate cross-species correspondences that are supported by genomic synteny and consistency in embedding space. This formulation can be viewed as orthology-aware annotation transfer, where predicted loci represent plausible functional counterparts rather than definitive orthologs.

1.2 The ncRNA orthology challenge

Inferring orthology for ncRNAs remains substantially more difficult. Sequence conservation is often weak or absent across evolutionary distances[8–13], and transcript boundaries are frequently ambiguous, making it unclear what constitutes a conserved unit[8–10,14]. Functional constraints are often not well captured by primary sequence alone, limiting the applicability of sequence-based

approaches[11]. In addition, ncRNA annotations are incomplete and heterogeneous across species[14,15], complicating both evaluation and transfer. Finally, embedding representations of short ncRNAs are context-dependent[16], making direct comparison sensitive to boundary definition.

For example, two ncRNA loci may exhibit low primary sequence similarity while preserving a similar structural pattern:

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GAGGUAUCCAAGAUACCUC
(((((((.....)))))))
AUGCUACGAUCGGUAGCAU
(((((((.....)))))))
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In such cases, sequence-based comparison may fail to establish correspondence, whereas embedding-based representations may capture similarities associated with conserved structural patterns.

1.3 Limitations of existing approaches

Existing approaches address parts of this problem but do not provide a scalable framework for ncRNA orthology inference. Covariance models (e.g., Infernal[17]/Rfam[18]) offer high specificity but rely on curated families and do not generalize to genome-wide discovery. Structure-based alignment methods such as LocARNA[19] and RNAforester[20] capture structural similarity but are computationally expensive and not suitable for large-scale analysis. Genome-wide discovery or annotation methods such as RNAz[21], EvoFold[22], and FEELnc[23] can identify candidate RNA elements or lncRNA loci, focusing on detection or annotation rather than orthology-resolved transfer. As a result, existing methods typically either lack scalability, require strong prior knowledge, or do not establish orthology-aware correspondence between loci at genome scale. While comparative genome alignments define candidate homologous intervals, they do not by themselves provide an RNA-specific measure of correspondence or a principled way to localize conserved structured subregions[24,25].

1.4 Synteny-constrained embedding matching

A key missing component is the integration of synteny with structure-aware similarity. Genome alignment chains define a constrained search space of candidate orthologous loci, reducing the combinatorial complexity of cross-species comparison[24,25]. At the same time, RNA foundation models such as RNA-FM[16], RiNALMo[26], and related embedding approaches such as RNABERT[27] provide learned representations that have been shown to encode information associated with RNA structure, family identity, and downstream functional prediction[28]. Without synteny constraints, embedding-based matching would require genome-wide comparison over a vastly larger search space, making the problem computationally expensive and increasing the risk of false positives. Combining these sources of information enables tractable comparison of ncRNA loci even when sequence conservation is limited.

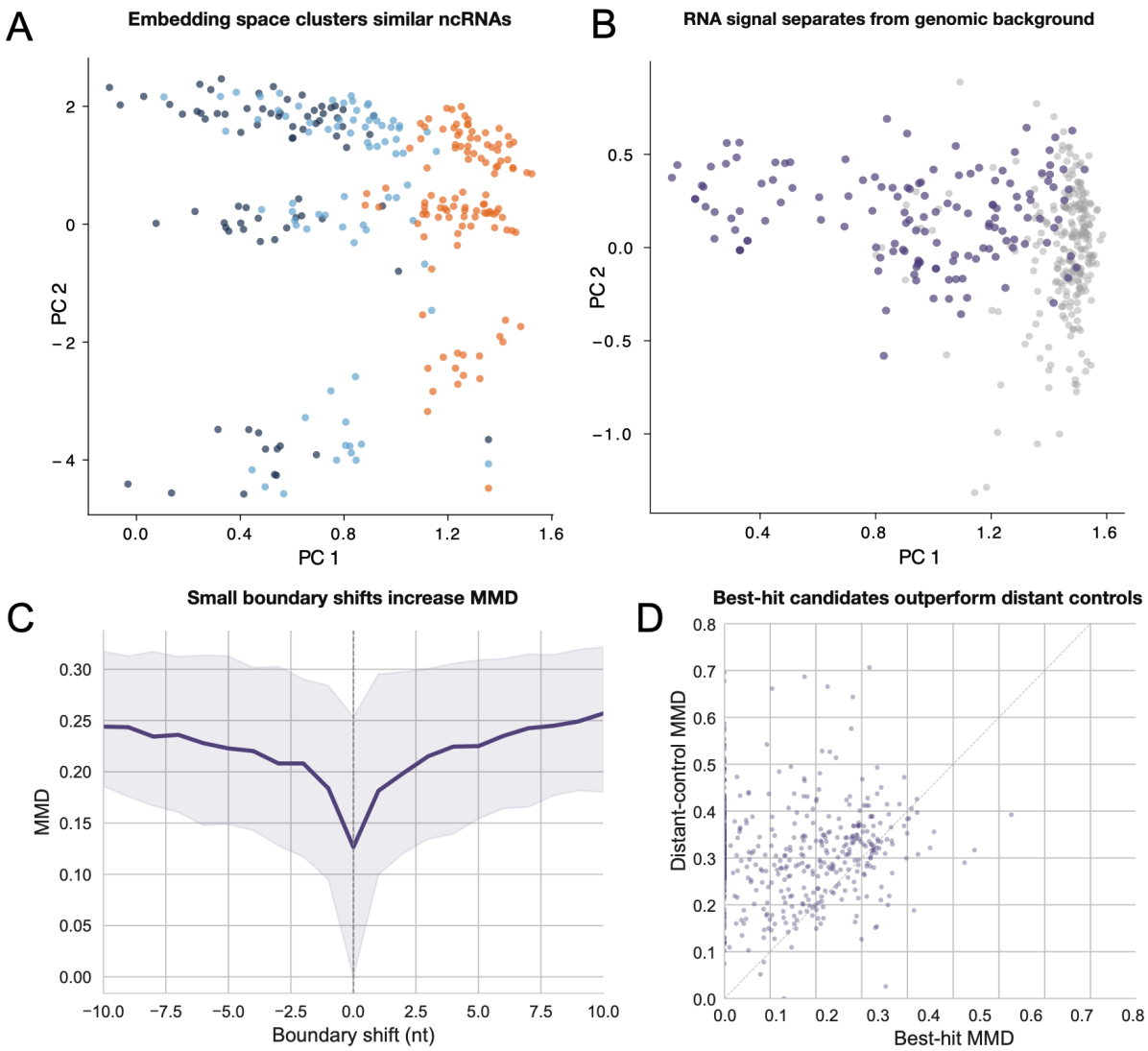

Figure 1. Embedding representations capture structure-aware similarity of ncRNAs and support local matching.
(A) Per-nucleotide RNA-FM embeddings (PCA) for two tRNA copies (blue) and a miRNA (orange). Each point corresponds to a nucleotide. tRNA embeddings cluster together, while the miRNA forms a distinct cluster.
(B) Mean-pooled embeddings for annotated ncRNAs (purple) and background sequences (gray; intergenic and dinucleotide-shuffled). Each point represents one sequence. RNA-derived sequences separate from genomic background despite reduced variance after pooling.
(C) MMD as a function of boundary shift (±10 nt) for predicted loci. Similarity is minimized at the original position and degrades with positional perturbation.
(D) Best-hit MMD compared to a non-overlapping control region (160–320 nt away) within the same locus. Most points lie above the diagonal (85%), indicating that best-supported matches are more similar than nearby alternatives.

1.5 Method overview

We formulate ncRNA orthology inference as constrained matching of structure-bearing regions in embedding space, guided by genome alignment chains. Rather than relying on sequence similarity alone, candidate loci are identified using a TOGA-inspired projection that leverages both syntenic

consistency and coverage of neutrally evolving sequence (e.g., intronic and flanking regions), which remains more reliably alignable across species[1,2].
CURIA implements this formulation as a three-stage pipeline. First, genome alignment chains define candidate loci, restricting the search space. Second, structured subregions ("islands") are detected within transcripts using embedding-based classification, providing localized units of comparison for long ncRNAs. Third, similarity between reference and query regions is evaluated using a distribution-level measure in embedding space. For short ncRNAs, this results in a single refined locus, while for long ncRNAs correspondence is represented as a set of matched islands.
Each region is represented as a collection of per-position embedding vectors. Instead of aligning sequences directly, CURIA compares regions at the level of embedding distributions using Maximum Mean Discrepancy (MMD)[29]. Intuitively, MMD measures how similar two sets of embedding vectors are: low values indicate similar distributions and therefore similar underlying embedding distributions, potentially reflecting structural constraints, while higher values correspond to dissimilar or background signal.
This formulation avoids explicit sequence alignment and provides a continuous similarity score that can be used to rank candidate matches, making it suitable for ncRNAs where primary sequence conservation is weak.

1.6 Contributions

In this work, we:

Reformulate ncRNA orthology-aware ncRNA annotation as synteny-constrained matching in embedding space.

Introduce CURIA, a pipeline combining chain-based locus projection with embedding-based structure detection and comparison.

Propose an island-based representation for long ncRNAs to localize conserved signal.

Develop a boundary refinement strategy for short ncRNAs.

Demonstrate a genome-scale proof-of-concept across multiple mammalian genomes, showing that this approach enables orthology-aware ncRNA annotation beyond sequence conservation.

2. Methods

2.1 Pipeline overview

CURIA is a computational pipeline for genome-scale ncRNA orthology inference that combines chain-guided locus prediction with embedding-based structural comparison (Figure 2). Given a reference genome with ncRNA annotations and one or more query genomes linked by genome alignment chains, CURIA produces: (i) predicted orthologous loci, (ii) an embedding-based confidence score (MMD), and, for long ncRNAs, (iii) matched structured subregions ("islands").
The pipeline follows a branching architecture after an initial projection step, handling short and long ncRNAs separately. Short ncRNAs yield a single refined locus, whereas long ncRNAs yield a set of island-level alignments.
The pipeline consists of seven stages:

(1) Preprocessing. Overlapping isoforms are collapsed into a union transcript per locus.
(2) Orthologous region prediction. Genome alignment chains are used to identify candidate orthologous regions via a simplified TOGA-inspired classifier that combines chain-based synteny with coverage of neutrally evolving sequence (Section 2.3).
(3) Short ncRNA refinement. Candidate loci are refined using embedding-based local boundary search (Section 2.4).
(4) Reference island detection. Structured subregions ("islands") are detected within transcripts (Section 2.5).
(5) Chain-guided projection. Reference islands are projected to the query genome to define targeted search regions (Section 2.6).
(6) Query island detection. Candidate islands are identified in projected regions using the same procedure (Section 2.7).
(7) Island alignment. Reference and query islands are matched using MMD-based comparison (Section 2.8).

Outputs include orthologous region assignments, refined short ncRNA loci, and island-level alignments.

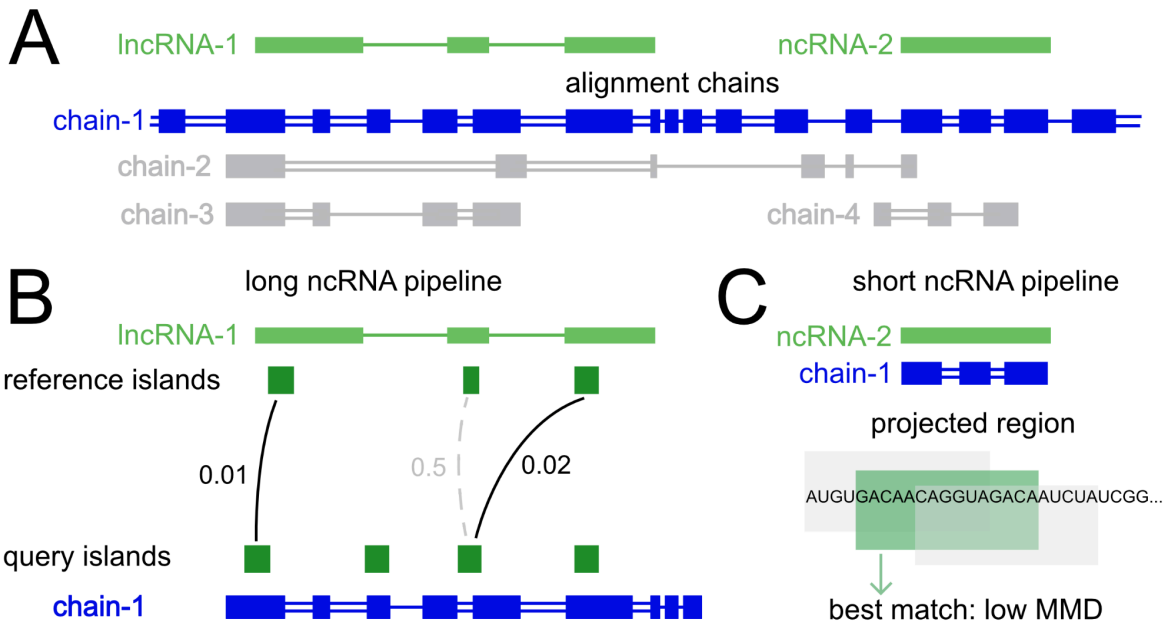

Figure 2. Overview of the CURIA pipeline for ncRNA orthology inference.
(A) Genome alignment chains restrict the search space by defining candidate orthologous loci between reference and query genomes.
(B) For long ncRNAs, structured subregions ("islands") are detected and matched across species using embedding-based similarity, enabling localization of conserved elements within transcripts.
(C) For short ncRNAs, orthology is resolved via local boundary refinement using sliding-window matching in embedding space, selecting the best-supported query region based on MMD.

2.2 Input data

CURIA requires three types of input data:

Reference genome and annotation. The reference genome is provided in 2bit format, with ncRNA annotations in BED12 format including transcript_id, gene_id, and biotype. Protein-coding transcripts are excluded.

Query genomes. Query genomes are provided in 2bit format. Query-side annotations are not required for prediction but may be used for evaluation.

Genome alignment chains. Pairwise genome alignment chains (UCSC chain format) define syntenic relationships between reference and query genomes. Low-scoring chains are filtered during loading. During preprocessing, overlapping isoforms are collapsed into union transcripts by merging exonic intervals. For lncRNAs, overlapping loci are further merged to reduce fragmentation.

2.3 Orthologous region prediction (RNA-TOGA)

Input. Union transcripts and genome alignment chains.
Operation. CURIA reuses the chain-guided locus projection strategy underlying TOGA[1], but replaces the original model with a simplified classifier tailored for ncRNA loci and used only to obtain coarse candidate regions rather than full transcript reconstructions (Figure 2A).
For each chain–transcript pair, a small set of features is computed:
(i) synteny, defined as the number of distinct genes intersecting the same alignment chain (log-transformed);
(ii) global exonic occupancy, defined as the fraction of aligned chain bases overlapping annotated exons;
(iii) flank coverage, defined as the fraction of ±10 kb flanking regions covered by chain alignment blocks.
A logistic regression classifier assigns an orthology probability based on these features. The classifier was trained on labeled chain–transcript pairs derived from human–mouse projections (details in Supplementary Methods). Pairs with predicted probability ≥ 0.5 were retained.
For each retained pair, candidate query intervals were obtained by projecting the transcript through the corresponding alignment chain. Intervals were defined as the span of chain blocks overlapping the transcript, including flanking regions to account for alignment imprecision. When multiple chains satisfied the threshold for a given transcript, all corresponding intervals were retained.
Output. Projected candidate regions with associated metadata.
Rationale. Only coarse locus-level projections are required to constrain downstream analysis, avoiding assumptions related to coding sequence structure and transcript reconstruction.

2.4 Short ncRNA pipeline

Short ncRNAs (e.g., miRNAs, tRNAs, snoRNAs, snRNAs) typically occupy compact loci, allowing direct local comparison without island-based decomposition (Figure 2C). This branch processes single-exon ncRNAs up to 160 nt. For each locus, the reference sequence is embedded using RNA-FM. The corresponding query region is extended symmetrically to account for alignment imprecision. Because embeddings are context-dependent[16,27], CURIA performs local boundary search. Candidate windows matching the reference length are tiled across the query region and compared to the reference using MMD (Section 2.8). The best match is refined by local boundary perturbation. Embeddings are projected from 640 to 16 dimensions using PCA prior to comparison (Section 2.10). Output: refined query coordinates with an MMD-based confidence score.

2.5 Reference lncRNA island detection

To detect RNA-like signal, we trained a logistic regression classifier on fixed-length sequence windows represented in RNA-FM embedding space. Positive examples consisted of 2,500 human ncRNA-derived windows (including lncRNAs, miRNAs, snoRNAs, snRNAs, scaRNAs, and misc_RNAs), while negative examples consisted of 2,500 intergenic, non-exonic genomic windows. All sequences were clipped to 72 nt, embedded with RNA-FM, mean-pooled across positions, and projected from 640 to 16 dimensions using PCA prior to training. The classifier was trained on a balanced dataset with a standard 80/20 train/test split.

The resulting model distinguishes embedding patterns associated with annotated ncRNAs from genomic background and is used as a lightweight detector of localized signal rather than a general ncRNA annotation model [19,20].

Each reference union transcript is scanned using a sliding window (72 nt, stride 16 nt). Windows are embedded, projected to PCA space, and scored by the classifier. The resulting probability trace is smoothed using a moving average filter (window size = 5), and contiguous regions exceeding a fixed probability threshold (0.25) are defined as islands.

Island coordinates are mapped to genomic space, allowing multi-segment islands across splice junctions.

Output. Reference islands with genomic coordinates and associated scores.

2.6 Chain-guided projection of islands

Input. Reference islands, genome alignment chains, and orthologous region assignments.

Reference islands are projected through alignment chains to define candidate query regions (Figure 2A). Intervals are extended to account for alignment imprecision.

Projected intervals on the same chromosome and strand are merged into non-redundant regions while retaining links to the originating reference islands.

Output. Query regions with mappings to reference islands.

Rationale. Projection restricts downstream search to compact regions while preserving relevant structural signal.

2.7 Local query island evaluation

Input. Merged query regions, query genome, and the classifier from Section 2.5.

Each query region is scanned using the same procedure as reference island detection. Windows are embedded, projected, and scored, and contiguous high-scoring regions are defined as islands.

Basic filtering removes regions with implausible size or lacking corresponding reference signal.

Output. Query islands with genomic coordinates and scores.

2.8 Island alignment in embedding space

Window similarity is quantified using Maximum Mean Discrepancy (MMD) with an RBF kernel, treating embeddings as empirical distributions:

$$MMD^2(X,Y) = \frac{1}{n(n-1)}\sum_{i\neq j} k(x_i,x_j) + \frac{1}{m(m-1)}\sum_{i\neq j} k(y_i,y_j) - \frac{2}{nm}\sum_{i,j} k(x_i,y_j)$$

where k(x, y) = exp(-γ ||x - y||^2).

Here, X = {x_1, ..., x_n} and Y = {y_1, ..., y_m} denote the sets of per-position embedding vectors for two sequence windows. The kernel bandwidth γ is set using the median pairwise distance heuristic computed on embedding vectors within each comparison.
Reference and query islands are decomposed into fixed-length windows (96 nt, stride 4 nt). For each pair of windows, MMD is computed, yielding an alignment matrix of size (n_windows_ref x n_windows_query).
To extract local correspondences, the MMD matrix is converted into a similarity matrix using a linear transformation S = tau - MMD, where tau is a constant offset (empirically set to 0.15). Local alignments are then extracted using a Smith-Waterman procedure[30] with affine gap penalties (gap open = 0.03, gap extend = 0.01).
Candidate matches are filtered using the following constraints: (i) minimum alignment length >= 40 nt; (ii) mean MMD <= 0.15; (iii) collinearity: the largest non-crossing subset of matches is retained via longest increasing (same strand) or longest decreasing (opposite strand) subsequence selection.
To avoid redundant matches, each query island is assigned to at most one reference island using a greedy selection procedure that prioritizes lowest mean MMD.
MMD values are reported as distance scores for downstream ranking and filtering of candidate matches.

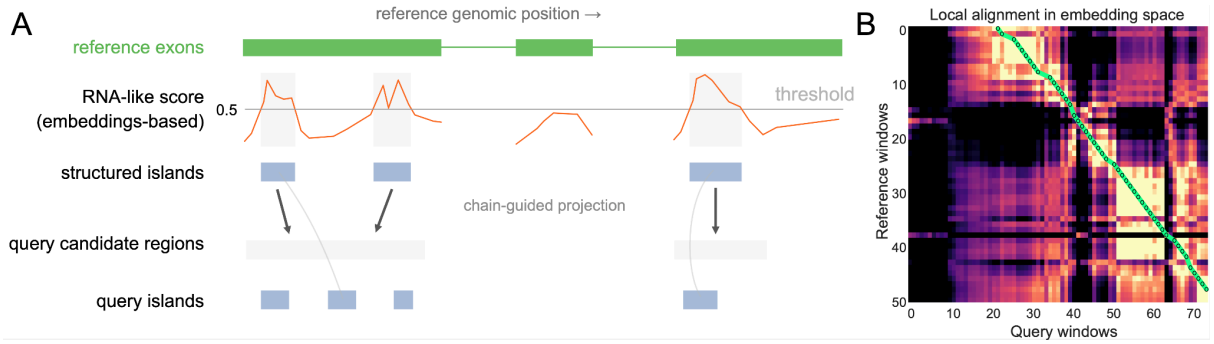

Figure 3. Identification and matching of structured ncRNA subregions using embedding-based similarity. (A) RNA-like signal along a reference transcript is detected using an embedding-based score, and thresholded to define structured subregions ("islands"). These islands are projected via genome alignment chains to define candidate regions in the query genome, where corresponding subregions are identified. (B) Similarity between reference and query regions is computed as an MMD-based alignment matrix, and local matches are extracted using a Smith–Waterman procedure.

2.9 Evaluation metrics

Evaluation is based on agreement with existing annotations and is used as an approximate proxy for ground truth. Due to incomplete and heterogeneous annotation of ncRNAs across species, these metrics should be interpreted as measures of annotation agreement rather than true biological accuracy.

2.9.1 Short ncRNA evaluation

For short ncRNAs, evaluation is performed at the locus level using overlap with annotated transcripts. A predicted region is considered a match if it overlaps an annotated ncRNA with ≥50% reciprocal overlap. We report:

- **annotation agreement**: fraction of predictions overlapping annotated loci

- **locus recovery**: fraction of annotated loci recovered by predictions
- **near-complete matches**: fraction of predictions with ≥99% overlap

These measures assess how well predicted loci align with annotated boundaries, while accounting for minor positional variation.

2.9.2 lncRNA locus-level evaluation

For lncRNAs, evaluation is performed at the locus level with relaxed criteria due to boundary ambiguity. A predicted locus is considered supported if it overlaps any annotated exon of an ncRNA gene.

We report:

- fraction of loci overlapping annotated exons
- distribution of overlap proportions across predictions

This reflects coarse agreement with annotated loci rather than precise boundary recovery.

2.9.3 Island-level evaluation

For long ncRNAs, evaluation is extended to structured subregions ("islands"). A matched island is considered supported if it overlaps any annotated ncRNA exon in the query genome.

We report:

- fraction of matched islands overlapping annotated regions
- number of matched islands per locus
- cross-species consistency of island matches

This analysis captures conservation at the substructure level, independent of full transcript recovery.

2.9.4 Annotation agreement vs confidence

To assess the relationship between embedding-based similarity and annotation support, we analyze annotation agreement as a function of MMD.

Predictions are stratified by MMD values, and we report:

- fraction of predictions overlapping annotations across MMD bins
- distribution of MMD values for overlapping and non-overlapping predictions

This provides an empirical characterization of MMD as a confidence measure, rather than relying on predefined thresholds.

2.9.5 Limitations

Predictions outside annotated regions may represent either false positives or unannotated ncRNAs[10,14,15]. Annotation completeness varies across species, and ncRNA boundaries are often

imprecise. As a result, all reported metrics reflect agreement with current annotations and should be interpreted as a lower bound on performance.

2.10 Implementation details

Embeddings are projected from 640 to 16 dimensions using PCA. The PCA transformation was precomputed on a large corpus of RNA-FM embeddings (~10^6 sequence positions sampled across diverse human transcripts) to provide a stable, dataset-independent representation.
Embedding inference is performed on GPU using batched execution, whereas downstream MMD computation, alignment, and filtering are performed on CPU. Internally, CURIA is organized as a staged task pipeline that separates embedding inference from downstream analysis. Sequence windows are processed by a shared GPU worker using micro-batching, while subsequent matching and filtering steps operate on persisted intermediate results. This design improves hardware utilization, supports incremental execution, and allows reference-side preprocessing to be reused across query genomes.
Outputs are reported in standard tabular and BED formats. The use of fixed embedding models and PCA projections ensures consistent representations across runs.

3. Results

3.1 Benchmark setup

CURIA was evaluated on a panel of mammalian genomes, using human (hg38) as the reference and multiple vertebrate genomes as queries (rhesus macaque (rheMac10), mouse (mm39), rat (rn7), cow (bosTau9), pig (susScr11), horse (equCab3), cat (felCat9), hedgehog (eriEur2), armadillo (dasNov3), and opossum (monDom5)). Genome assemblies and ncRNA annotations were obtained from GENCODE (human v48, mouse VM37).

Across species, on the order of ~20,000 ncRNA transcripts are annotated per genome. These counts reflect available annotations rather than true transcript repertoires and include both well-supported elements (e.g., tRNAs, rRNAs, selected snoRNAs) and less-characterized or potentially spurious transcripts, particularly among lncRNAs.

Evaluation is therefore interpreted as agreement with existing annotations rather than direct measurement of biological accuracy. Due to incomplete and heterogeneous annotation, reported results should be considered a lower bound on true performance.

Evaluation was performed at both the locus level (for short ncRNAs and lncRNA loci) and the substructure level (for island matches), following the criteria defined in Section 2.9. Absence of overlap with annotated regions does not necessarily indicate a false positive.

3.2 Overall performance

We first evaluated agreement between predicted loci and existing annotations. Among 2,805 predicted short ncRNA loci, 53.1% overlap annotated exons, with 47.7% covering ≥50% of the annotated transcript and 35.5% showing near-complete overlap (≥99%). Among overlapping predictions, the median overlap is 100% (mean 88.8%), indicating that matched loci are typically well localized and consistent with annotated boundaries.

These results suggest that localization of matched loci is generally accurate when annotation support is present. The substantial fraction of predictions without annotation overlap is consistent with a combination of annotation incompleteness, projection errors, and potential false positives. Notably, many non-overlapping predictions are supported by strong synteny, suggesting that some may correspond to unannotated or weakly annotated ncRNA loci.

3.3 Short ncRNA performance and MMD behavior

We evaluated similarity between matched short ncRNA loci using MMD in embedding space. Across all predictions, the MMD distribution is skewed toward low values (median 0.108; Fig. 4A), with a substantial fraction of near-zero values, indicating consistent embedding representations between reference and query loci.

Stratification by biotype reveals clear differences (Fig. 4.C). snoRNAs and miRNAs exhibit the lowest MMD values, consistent with their compact size and well-defined structural constraints. In contrast, snRNAs and micsRNA show higher and broader MMD distributions, likely reflecting increased ambiguity due to copy number, sequence divergence, and genomic context. tRNAs serve as a positive control, with consistently low MMD values in line with their well-characterized structural conservation[31].

MMD is strongly associated with annotation agreement. Predictions with low MMD values (≈ <0.1) show high overlap with annotated loci (~90%), whereas higher MMD values are associated with reduced overlap and increased dispersion. Predictions with MMD >0.2 rarely overlap annotations and are enriched for ambiguous cases. These ranges are empirical and used for interpretation rather than as fixed thresholds.

Consistent with this, loci overlapping annotations are enriched at low MMD values, while non-overlapping predictions are shifted toward higher values (Fig. 4.B), supporting the interpretation of MMD as a continuous confidence measure.

Notably, a subset of high-confidence predictions (MMD ≈ 0) does not overlap existing annotations despite strong syntenic support, suggesting that incomplete annotation may contribute substantially to the observed gap, although projection errors and false positives are also likely to contribute.

To assess whether embedding-based similarity merely recapitulates sequence conservation, we compared MMD to pairwise sequence identity for matched short ncRNA loci. The two measures were strongly anticorrelated (Pearson $r = -0.797$; Spearman $\rho = -0.846$; $n = 2{,}697$), indicating that low MMD generally coincides with higher primary-sequence similarity. However, the relationship was not absolute: eight loci showed low MMD (<0.1) despite sequence identity below 50%, consistent with a residual signal not fully explained by pairwise sequence identity alone.

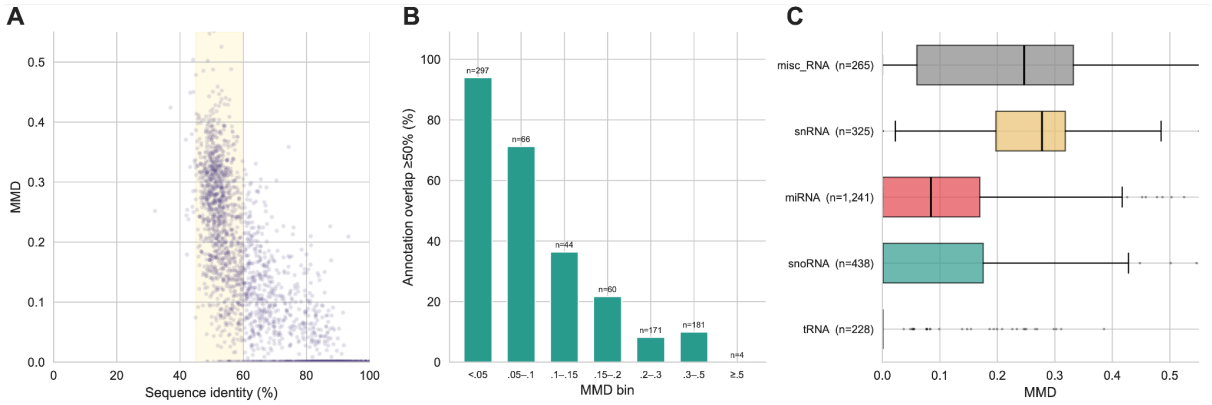

Figure 4. Behavior of embedding-based similarity (MMD) across ncRNA loci.

(A) Relationship between sequence identity and MMD for matched loci. While the two measures are strongly anticorrelated, substantial variability is observed at intermediate sequence identity (~45–60%), indicating that embedding-based similarity captures signal beyond primary sequence.
(B) Annotation agreement as a function of MMD. The fraction of loci with ≥50% overlap with annotated ncRNAs decreases with increasing MMD, supporting its interpretation as a continuous confidence measure.
(C) Distribution of MMD values across ncRNA biotypes. Compact RNAs (miRNAs, snoRNAs, tRNAs) exhibit low and narrow distributions, whereas snRNAs and other RNA classes show broader and higher MMD values.

3.4 Structural conservation analysis ("islands")

We analyzed structured subregions ("islands") within lncRNAs to assess conservation beyond locus-level correspondence. Island detection reveals that structured signal is sparse and localized. Most lncRNAs contain a small number of discrete islands rather than uniform signal across the full transcript, as illustrated by representative loci (Fig. 6A), indicating that the conserved signal is concentrated in compact subregions.
Across loci, islands were detected in approximately 74% of lncRNAs, with a median of 2 islands per transcript. The distribution is skewed, with most transcripts containing one or two islands and a minority exhibiting more complex patterns.
To assess conservation, we evaluated cross-species matches between reference and query islands. At the gene level, 83.9% of lncRNAs exhibit at least one cross-species match, indicating that conserved signal is widespread but rarely spans entire transcripts. Consistent with this, conservation at the level of structured subregions is partial: 83% of cores are recovered in at least two species, 49% in five or more species, and only ~1% are consistently detected across all species, indicating a spectrum of conservation from weakly shared to highly stable elements.
Only 22% of matched islands overlap annotated ncRNA regions in the query genome, and 17% show ≥50% exonic overlap, suggesting that many conserved candidate elements are not captured by current annotation resources.
Conservation patterns are heterogeneous across loci. Some lncRNAs exhibit a small number of stable, reproducible matches across species, whereas others show fragmented or species-specific signal. Together, these observations support a model in which lncRNA conservation is modular, with signal preserved in discrete structured regions rather than across entire transcripts[10–12,14,32].

3.4.1 Multi-species conserved cores

To identify conserved structural elements, we examined islands that are consistently recovered across multiple species. We observe a subset of islands that exhibit reproducible matches across species with low MMD values (Fig. 6B–C). These conserved cores represent candidate conserved elements maintained despite sequence divergence.
Across all detected cores, 83% are recovered in at least two species, 49% in five or more species, and approximately 1% are consistently detected across all ten species, including the marsupial outgroup (opossum), spanning deep evolutionary divergence (Fig. 6B), indicating a spectrum of conservation ranging from weakly shared to highly stable elements.
In contrast to locus-level conservation, which is often incomplete or ambiguous, conserved cores exhibit stable cross-species correspondence at the level of localized embedding similarity. They are frequently detected even in loci where full-transcript correspondence is weak or absent, reinforcing the modular view of lncRNA conservation.
To assess whether such matches could arise by chance, we compared results to a randomized baseline in which query islands were replaced with randomly sampled regions within the same syntenic

intervals. Under this null model, consistent low-MMD matches were rare, indicating that such matches are not readily reproduced under randomized controls within the same syntenic intervals, but reflect reproducible cross-species similarity in embedding space.

3.5 Candidate conserved regions absent from current annotations

We next assessed the prevalence of such candidate elements across loci. After excluding regions overlapping or proximal to protein-coding genes, we identify approximately 700 intergenic loci exhibiting consistent low-MMD similarity in embedding space across ≥9 species. The full set of loci is available in Data and Code Availability (preprint_results/lncRNAs_with_conserved_cores.txt). MMD exhibits an empirical separation regime, with the majority of high-confidence matches falling below ~0.1, beyond which few additional loci are recovered. This suggests a practical regime for identifying reproducible signal in embedding space. The persistence of these loci after filtering suggests that the observed signal is unlikely to be primarily driven by coding-associated conservation. Together, these results suggest that such candidate conserved elements are not rare and extend beyond currently annotated ncRNAs.

3.6 Case studies

To illustrate the behavior of CURIA in representative scenarios, we examined individual loci, including (i) well-supported conserved ncRNAs, (ii) cases with incomplete or ambiguous annotation, and (iii) loci with structural variation (Fig. 5A–C).

3.6.1 Clean reference-supported match

We first consider a well-characterized ncRNA locus with consistent annotation across species (Fig. 5A). CURIA recovers a precise correspondence between human SNORD57 and mouse Snord57, with accurate localization and full agreement with existing annotations. This example demonstrates reliable recovery of conserved ncRNA loci under strong structural constraint.

3.6.2 Recovery under incomplete or ambiguous annotation

In some cases, CURIA identifies loci that overlap weakly annotated or ambiguous regions (Fig. 5B). For example, projection of human RNU6-7 recovers a corresponding mouse locus (Gm24859), despite limited annotation support. This illustrates that CURIA can recover biologically consistent loci even when annotation is incomplete.

3.6.3 Length divergence and partial correspondence

Finally, we consider a case where transcript structure differs between species (Fig. 5C). Projection of human vault RNA identifies a corresponding region overlapping mouse Vaultrc5, despite differences in transcript length. CURIA localizes the most similar segment within the syntenic region, indicating that correspondence is driven by localized structural similarity rather than full-length transcript conservation.

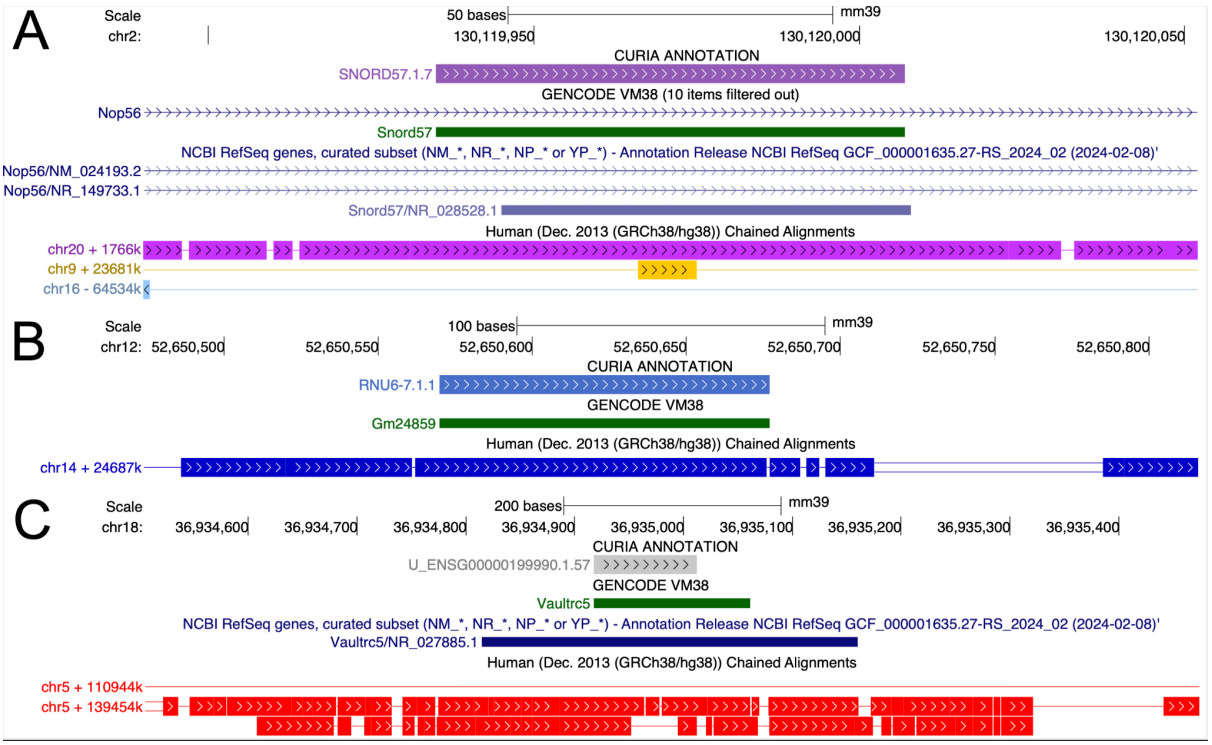

Figure 5. Recovery of short ncRNA loci across species using CURIA.
Genome browser snapshots (UCSC) illustrating representative examples of short ncRNA projection from human (hg38) to mouse (mm39). Tracks show CURIA predictions, GENCODE annotations, and alignment chains.
(A) Projection of human SNORD57 to mouse identifies a nearly exact match overlapping annotated Snord57, demonstrating precise localization under strong structural conservation. (MMD = 0.0)
(B) Projection of human RNU6-7 intersects an annotated mouse locus (Gm24859), indicating recovery of a corresponding genomic region despite ambiguous or incomplete annotation. (MMD = 0.0)
(C) Projection of human vault RNA (ENSG00000199990) identifies a corresponding locus overlapping mouse Vaultrc5. The matched region differs in length from the human transcript, yet CURIA localizes the most similar segment within the syntenic region. (MMD = 0.059)

3.7 Error analysis

We analyzed failure cases to identify the primary sources of error in CURIA predictions. Errors arise from limitations in chain-based projection, incomplete reference annotation, weak embedding signal, and modeling assumptions.

Chain-based projection defines the search space and represents a hard constraint: if the orthologous region is not captured by genome alignments, it cannot be recovered. In addition, duplicated or paralogous regions may lead to ambiguous or incorrect projections.

CURIA also depends on reference ncRNA annotations, which are incomplete and often imprecise, particularly for lncRNAs. Missing or truncated annotations limit the recoverable signal and can result in partial matches rather than full-length correspondence.

A further limitation is weak or diffuse structural signal. Some loci lack clearly detectable islands in embedding space, leading to ambiguous or fragmented matches. In addition, approximate projection and context-dependent embeddings can affect boundary localization, particularly for short ncRNAs or compact conserved regions.

Finally, the method assumes approximate collinearity and relies on local similarity, and therefore does not capture rearrangements of structured elements or long-range RNA interactions. CURIA also does not perform de novo discovery or assess transcriptional activity.

Overall, errors are more often attributable to missing or ambiguous signal, annotation limitations, or projection constraints than to failures of the matching procedure itself.

3.8 Runtime and scalability.

We evaluated CURIA on a representative genome pair (hg38 vs mm39). Using an AMD EPYC 9575F CPU and an NVIDIA RTX PRO 6000 GPU, the full pipeline completes in ~30 minutes when restricted to primary isoforms. For a more comprehensive transcript set, runtime is expected to remain within ~1 hour.
Runtime is dominated by island matching, while upstream steps, including orthologous region prediction and short ncRNA refinement, complete within 20–30 minutes. Reference and query island detection each require ~30 minutes and are restricted to syntenically projected regions.
Runtime scales primarily with the total exonic sequence in the reference annotation rather than genome size. The number of islands per locus is relatively stable, and no combinatorial explosion is observed, although loci with complex island structure may incur higher cost.
Embedding inference is performed on GPU with batched execution (up to 4096 sequences per batch), while MMD computation and alignment are CPU-bound and represent the primary bottleneck. Individual island matching jobs typically complete in under one second, but dominate runtime in aggregate.
Overall, CURIA scales approximately linearly with annotation size and can be applied to multi-species panels without prohibitive computational cost given a single modern GPU.
The observed runtime is enabled in part by decoupling GPU-bound embedding inference from CPU-bound matching and by amortizing inference overhead through micro-batching of sequence windows.

4. Discussion

CURIA should be viewed primarily as a proof of concept demonstrating that synteny-constrained embedding comparison yields biologically meaningful signal for ncRNA correspondence at genome scale, rather than as a finished annotation framework optimized for exhaustive sensitivity or production use.
Within this framing, several key observations emerge.

4.1 Key findings

We show that ncRNA orthology inference can be formulated as synteny-constrained matching in embedding space. By combining genome alignment chains with structure-aware embeddings, CURIA enables cross-species comparison of ncRNA loci even in the absence of strong sequence conservation. Different components of the method play complementary roles: synteny restricts the search space, while embedding-based similarity provides the primary discriminative signal. For long ncRNAs, decomposition into structured subregions (“islands”) further localizes conserved signal and enables tractable genome-scale comparison.
Our results indicate that meaningful cross-species correspondence can be recovered despite weak sequence identity, suggesting that embedding representations capture structural features relevant for ncRNA conservation.

4.2 Comparison to existing methods

Existing approaches address complementary aspects of ncRNA detection and comparison but do not provide a scalable framework for genome-wide orthology inference.
Covariance model–based methods (e.g., Infernal/Rfam) achieve high specificity by modeling sequence and secondary structure but rely on curated families and do not generalize to genome-wide discovery. Structure alignment methods (e.g., LocARNA[19], RNAforester[20]) capture structural similarity but are computationally expensive and limited to small candidate sets. Genome-wide scanning approaches (e.g., RNAz, EvoFold, FEELnc) identify candidate RNA elements but do not establish explicit cross-species correspondence.
In contrast, CURIA combines synteny with embedding-based similarity to enable tractable genome-scale inference of ncRNA correspondence without requiring curated families or explicit structural alignment.

4.3 Biological implications

Our results suggest that conservation in lncRNAs is often heterogeneous and not uniformly distributed across full transcripts. While sequence similarity and embedding-based similarity are strongly correlated for short ncRNAs (Fig. 4A; Pearson $r = -0.80$; Spearman $\rho = -0.85$), substantial variability in MMD is observed at intermediate sequence identity levels (≈50–60%), indicating that sequence conservation alone does not fully capture cross-species correspondence. Instead, conservation frequently manifests as localized conserved elements. A small number of loci show low MMD despite sequence identity below 50%, consistent with a residual signal not fully explained by sequence similarity alone.
Only subsets of transcripts exhibit consistent cross-species correspondence, while other regions show weak or no detectable similarity, supporting a model of modular organization with distinct evolutionary constraint[10,12,14].
The island-based representation provides a natural way to capture this behavior: conserved islands correspond to candidate conserved cores, while intervening regions appear less constrained. This is consistent with previous observations that lncRNAs often lack strong global sequence conservation yet retain localized conserved features.
Importantly, embedding-based similarity is not fully reducible to sequence identity. While low MMD generally coincides with higher sequence similarity, regions with comparable sequence identity can exhibit substantially different MMD values, suggesting sensitivity to features beyond primary sequence, potentially including structural or contextual signals.
Conservation patterns also vary across RNA classes. Compact RNAs such as tRNAs and snoRNAs tend to exhibit strong and consistent correspondence, whereas long ncRNAs frequently show partial or fragmented conservation, suggesting that correspondence is more localized and less uniformly defined in longer transcripts.
Overall, these observations support a model in which conserved candidate elements are embedded within more weakly constrained transcript contexts, emphasizing the importance of localized, structure-aware comparison for cross-species analysis.

Figure 6. Modular and heterogeneous conservation of lncRNA cores across species.
(A) Cross-species matching of structured cores for representative lncRNAs (NORAD, PVT1, FIRRE). Rows correspond to species and columns to reference cores ordered along the transcript; color indicates MMD, and gray denotes absence of a supported match. Core positions along the reference transcript are shown below each heatmap.
(B) Distribution of core reproducibility across species, shown as the fraction of cores detected in ≥k species.
(C) Median island-level MMD across species (± IQR), showing a consistent increase with phylogenetic distance.

4.4 Limitations

This study has several limitations related to data availability, modeling assumptions, and evaluation. First, the lack of reliable ground truth for ncRNA orthology remains a major challenge. Existing annotations are incomplete and heterogeneous across species, and transcript boundaries—particularly for lncRNAs—are often ambiguous. As a result, evaluation based on annotation overlap provides only an approximate and conservative estimate of performance.
Second, CURIA depends on the availability and quality of genome alignment chains. While chains effectively constrain the search space, errors or fragmentation in alignments can affect downstream predictions. In addition, the orthology classification step is adapted from protein-coding genes and may not fully capture the diversity of ncRNA conservation patterns.
Third, the method relies on embedding representations derived from RNA foundation models. Although RNA-FM provides a practical representation, it may not capture all relevant structural features. In particular, CURIA is primarily sensitive to localized structure and does not model long-range interactions or global RNA architecture.
Fourth, CURIA does not establish transcriptional activity or molecular function. Predicted regions represent candidate conserved elements and require additional experimental or transcriptomic validation.
Fifth, absence of overlap with annotated loci does not necessarily indicate false positives, and reported performance likely underestimates true recall due to annotation incompleteness.

Finally, CURIA does not explicitly distinguish between homology and analogy at the level of embedding similarity. While synteny strongly biases predictions toward orthologous loci, embedding similarity may also capture convergent structural features or recurrent local motifs. Accordingly, high-confidence matches should be interpreted as candidate orthologous elements supported by both synteny and structural similarity, rather than definitive evidence of shared evolutionary origin.

4.5 Embedding-based similarity and interpretability

Embedding representations used in CURIA are derived from RNA foundation models trained on large collections of RNA sequences without explicit structural supervision[16,27,28]. Prior work has shown that such models capture information associated with RNA secondary structure and related functional signals. Consistent with this interpretation, RNA language model representations have also proven useful in downstream RNA structure prediction tasks[33].

In this setting, embedding similarity provides an implicit measure of correspondence in embedding space, plausibly influenced by structural features. Rather than comparing sequences directly, CURIA compares distributions of per-position embeddings using MMD, enabling similarity to be assessed at the level of local patterns in embedding space.

Empirically, MMD values form a smooth empirical scale separating high-confidence matches from background signal. Low MMD values are associated with stronger cross-species correspondence and high overlap with annotated loci, whereas higher values correspond to weak or ambiguous matches.

This behavior varies across ncRNA classes. Compact and structurally constrained RNAs (e.g., miRNAs, snoRNAs) exhibit low MMD values and clear separation from background, whereas longer or less structured ncRNAs show broader, overlapping distributions. This suggests that embedding-based similarity reflects how consistently cross-species correspondence can be detected across ncRNA biotypes.

This idea is also consistent with findings from protein language models, where representations learned from sequence alone were shown to organize according to structural and functional properties, suggesting that unsupervised biological embeddings can encode signals not directly captured by primary sequence alone[34].

Overall, embedding-based similarity provides a continuous measure of correspondence confidence without requiring explicit structural alignment.

4.6 Failure modes

The observed failure modes reflect three primary limitations of the CURIA framework.

First, the method depends on the presence of localized structural signal. Loci lacking clearly detectable structure in embedding space often yield no identifiable islands or only weak, fragmented matches.

Second, CURIA relies on genome alignment chains to define the search space. Errors or fragmentation in alignments, as well as duplicated or paralogous regions, can result in missing or ambiguous projections.

Third, the approach assumes approximate collinearity and focuses on local similarity. As a result, it does not capture rearrangements of structured elements or long-range RNA interactions.

These limitations indicate that CURIA is most effective for loci with well-localized structural signal and reliable synteny, and that absence of a detected match does not necessarily imply absence of conservation.

5. Data and code availability

The CURIA pipeline is available at: https://github.com/kirilenkobm/CURIA_pipeline. Preprocessed reference annotation files used in this study are provided in the repository under `input_data/reference_annotation/`. All predicted annotations, intermediate outputs, and summary result files used in the analyses are available under `preprint_results/`, including the set of candidate conserved intergenic loci described in Section 3.5. Genome assemblies (2bit format) were obtained from NCBI and UCSC Genome Browser resources. Genome alignment chains were obtained from the TOGA project (Hiller lab): https://genome.senckenberg.de/download/TOGA. All external datasets are publicly available, and links to the exact sources are provided in the repository.

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